MILITARY SPECIFICATION

FIRING DEVICE, EXPLOSIVE ACTUATOR

(CCU-64/B)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the CCU-64/B Explosive Actuator Firing Device referred to herein as the device.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-85251	Bomb, Binary, Chemical, BLU-80/B.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1377

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification and Marking of U.S. Military Property.
MIL-STD-453	Radiographic Inspection.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-45662	Calibration System Requirements.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

SA2875117	Tail Fin Firing Device Test Fixture.
X4900606	Firing Device.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations

49 CFR 171-178 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

3.1 Item description. The device is a pyrotechnic device consisting of a housing, an impulse cartridge, and lanyard activated firing pin.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Tail fin deployment. When the tail fin lanyard is extracted at a rate of 20 ±2 feet per second with a pull force of 25 ±2 pounds, the device shall function the impulse cartridge after 10 ±1 inches of the lanyard are extracted. The explosive output shall be such that all four tail fins shall deploy and lock into position within 25 ±10 milliseconds (ms) after the device activates the impulse cartridge.

3.2.1.2 Tail fin safety. When the tail fin safety pin is installed and a pull force of 200 pounds is applied to the tail fin lanyard, the lanyard shall not extract nor shall the device be activated.

3.2.1.3 Lanyard extraction. When the tail fin safety pin is removed, the pull force on the tail fin lanyard required to rotate the spool shall not exceed 5 pounds.

3.2.2 Environmental. The device shall not be damaged nor its performance degraded after exposure to the following environments.

3.2.2.1 Temperature shock. The device shall withstand alternate sequential exposure to temperatures of -54 degrees Celsius (°C) and +74°C under the environmental conditions stated in 4.4.7.

3.2.2.2 Vibration. The device shall withstand exposure to the vibration spectrum specified in Figure 1 in each of three mutually perpendicular axes under the environmental conditions stated in 4.4.3.

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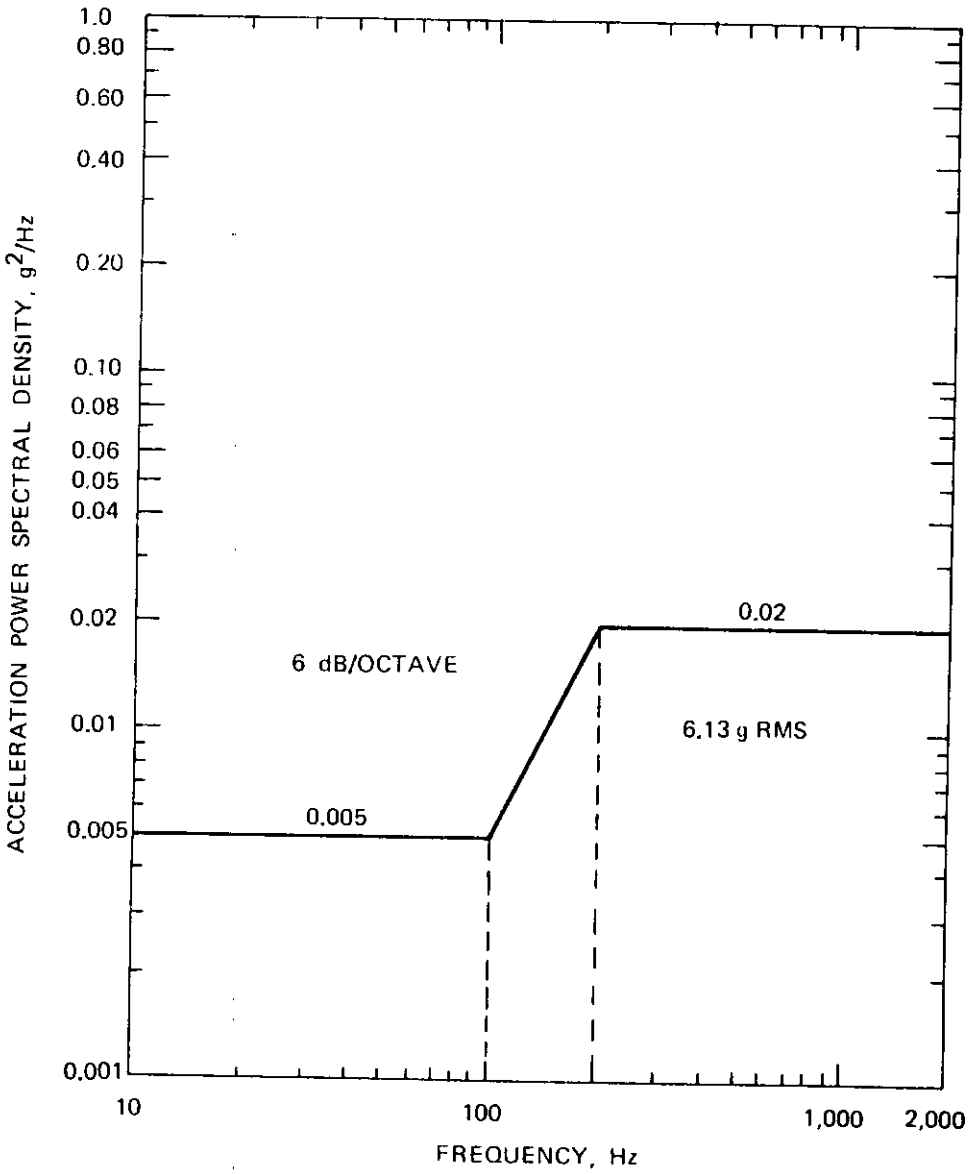


FIGURE 1. Vibration schedule.

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3.2.2.3 Mechanical shock. The device shall withstand exposure to half sine wave impact shocks having a peak amplitude of 15 gravity units (g) and a total duration of 11 ms under the environmental conditions stated in 4.4.6.

3.2.2.4 High temperature. The device shall withstand exposure to a high temperature of +74°C for 48 hours under the environmental conditions stated in 4.4.4.

3.2.2.5 Low temperature. The device shall withstand exposure to a low temperature of -48°C for 24 hours under the environmental conditions stated in 4.4.5.

3.3 Design and construction.

3.3.1 Production drawings. The device shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on Drawing X4900606 and as specified herein.

3.3.1.1 Mechanical integrity. The device shall show no evidence of defects and the internal construction shall conform to Drawing X4900606.

3.3.2 Standards of manufacture.

3.3.2.1 Certification. When specified in the contract or purchase order, the contractor shall include written certification to the procuring activity (see 6.2.2), accompanied by objective quality evidence as defined in MIL-STD-109, that the materials, processes, and parts used in the assembly meet the applicable requirements for each item.

3.3.2.2 Working environments. Any special working environments for the manufacture of the device shall be as specified in MIL-B-85251.

3.3.3 Marking. Each cartridge shall be marked in accordance with MIL-STD-130, with the manufacturer's name or symbol, part number and manufacturer's lot designation. The markings shall be as shown on Drawing X4900606.

3.4 First article. Unless otherwise specified in the contract or purchase order, a preproduction sample of 15 firing devices shall be delivered to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes and procedures proposed for production. Any production prior to acceptance of the preproduction

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sample is at the risk of the contractor. After approval of the preproduction sample, changes in material, parts, procedures, design, or processes, may be cause for resubmission of a preproduction sample.

3.4.1 Retest. At the discretion of the procuring activity, preproduction tests, or any portion thereof shall be repeated under any of the following conditions:

a. The manufacturer has modified the product (such as change of raw materials, processes, production procedures, or methods).

b. Where there is evidence that the quality of the product has not been maintained. This evidence may be in the form of accumulated product failures, of systems failures attributable to the product, or failure of the product to pass any of the tests for production lot acceptance that may be conducted by or for the procuring activity.

c. When applicable documents have been amended or revised sufficiently so that continued validity of the previous preproduction testing is questionable.

d. When the manufacturer has not produced firing devices to this specification for a period of 12 months or longer.

e. When the manufacturer changes the location at which explosive actuator firing devices are produced.

3.5 Documentation. When specified in the contract or purchase order (see 6.2.2), inspection reports shall be prepared as specified in 4.1.2.2 and 4.2.4.

3.6 Workmanship. The device shall be fabricated in a manner that will ensure compliance with all requirements of this specification and the assembly drawing. Unless otherwise specified herein or on the assembly drawing, fabrication and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspections. Preproduction inspections shall be conducted on the preproduction sample (see 3.4) and shall consist of the examinations and tests of Table I in the sequence shown.

4.1.2.1.1 Acceptance criteria. The failure of one or more devices to pass any of the preproduction tests shall cause rejection of the preproduction sample.

TABLE I. Preproduction inspections.

Examination or test	Requirement paragraph	Method paragraph
Group I (100%)		
Lanyard extraction	3.2.1.3	4.4.9
Visual/radiographic examination	3.3.1, 3.3.1.1, 3.3.6	4.4.1, 4.4.2
Group II		
High temperature	3.2.2.4	4.4.4
Low temperature	3.2.2.5	4.4.5
Temperature shock	3.2.2.1	4.4.7
Mechanical shock	3.2.2.3	4.4.6
Vibration	3.2.2.2	4.4.3
Functional	3.2.1.1, 3.2.1.2	4.4.8

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4.1.2.2 Preproduction inspection report. When specified in the contract or purchase order (see 6.2.2), results of the preproduction inspections shall be prepared for the procuring activity. The report shall contain the following information accompanied by a certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the preproduction sample complies with all quality assurance provisions of this specification for the device specified in the contract or purchase order.

b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Purchase descriptions or contract number and date together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Quality conformance inspections. The quality conformance inspections shall be performed on each inspection lot and shall consist of the examination and tests listed in Table II in the sequence shown.

4.2.1 Inspection lot definition. Inspection lot definition shall be in accordance with MIL-STD-105. An inspection lot shall be formed from devices which have passed Group I inspections.

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TABLE II. Quality conformance inspections.

Examination or test	Requirement paragraph	Method paragraph
Group I (100%)		
Lanyard extraction	3.2.1.3	4.4.9
Visual/radiographic inspection	3.3.1, 3.3.1.1, 3.3.6	4.4.1, 4.4.2
Group II (sample)		
Vibration	3.2.2.2	4.4.3
Functional	3.2.1.1, 3.2.1.2	4.4.8

4.2.2 Sampling. A sample shall be randomly selected from each inspection lot. The sample size shall be in accordance with MIL-STD-105 Normal Inspection, Inspection Level II, Single Sampling, and an Acceptable Quality Level (AQL) of 0.65 percent defective.

4.2.3 Acceptance criteria. Failure of one or more firing devices in the sample to pass any of the Group II quality conformance tests shall cause rejection of the inspection lot represented.

4.2.4 Quality conformance inspection report. When specified in the contract or purchase order (see 6.2.2), results of the quality conformance inspection shall be included with each lot. Reports shall contain the following information accompanied by a certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of this specification for the firing device specified in the contract or purchase order.

b. Number of units of product inspected.

c. Results obtained for all inspections performed.

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d. Contract or purchase order number and date together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3 Test equipment and conditions.

4.3.1 Standard test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the requirements of the contract or purchase order, specification, and drawing requirements at the rate of delivery specified in the contract or purchase order. The calibration of measuring and test equipment shall be in accordance with MIL-STD-45662. Test equipment, including associated calibration programs and operation and maintenance procedures are subject to disapproval by the procuring activity. When special test equipment and circuits are devised or commercially available equipment is employed, all test equipment circuits and methods are subject to disapproval by the procuring activity. Unless otherwise specified herein, the magnitude of any error introduced by test equipment shall not exceed 10 percent of the tolerance of the requirement being measured.

4.3.2 Test conditions. Unless otherwise specified herein, all tests shall be conducted at the test area at ambient temperature and relative humidity. Where special conditions of tests are specified, equipment employed shall adequately provide specified conditions, and all monitoring and measurement devices shall conform to the requirements of 4.3.1. When special conditions of tests are not imposed, commercially available direct measurement equipment may be employed provided they conform to the requirements of 4.3.1.

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4.4 Inspection methods.

4.4.1 Visual inspection. The device shall be visually inspected to determine conformance with the documents and drawings in accordance with 3.3.1. Firing devices not conforming to 3.3.1 shall be rejected.

4.4.2 Radiographic inspection. Radiographic inspection of the device shall be in accordance with MIL-STD-453. The device shall meet the requirements of 3.3.1.1 to be acceptable. Any device not meeting the requirements of 3.3.1.1 shall be rejected.

4.4.3 Vibration. Testing shall be conducted in accordance with MIL-STD-810, Method 514.2. Testing shall be conducted at a temperature of -48 ±3°C for 30 minutes per axis. The vibration test profile shall be in accordance with Figure 1. The tolerance on vibration test levels shall be in accordance with MIL-STD-810, Method 514.2, Random Vibration Tests.

4.4.4 High temperature. The device shall be subjected to the high temperature test of MIL-STD-810, Method 501.1, Procedure I, except the temperature shall be +74°C. Steps 4 and 5 shall be omitted.

4.4.5 Low temperature. The device shall be subjected to the low temperature test of MIL-STD-810, Method 502.1, Procedure I, except the temperature shall be -48°C. Steps 4 and 5 shall be omitted.

4.4.6 Mechanical shock. Testing shall be performed in accordance with MIL-STD-810, Method 516.2, Procedure I, with the following modifications or additions. Testing shall be conducted by applying three shocks in each direction through three mutually perpendicular principle axes at a temperature of:

a. The shock pulse signature shall be a half sine pulse, 15 g peak and 11 ms duration. The shock pulse tolerance limits shall be in accordance with MIL-STD-810, Method 516.2, Figure 516.2-2, "Half Sine Shock Pulse Configuration and its Tolerance Limits."

b. Before applying mechanical shocks, the device shall be temperature conditioned at -48 ±3°C for not less than 4 hours. If the device must be removed from the chamber to apply the required shocks, the elapsed time between removal from the chamber and the start of application of any shock along any axis may be increased to 5 minutes, provided the firing devices are not exposed to ambient temperature for more than

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3 minutes of the 5 minute period. If shock testing is not completed within the time limits specified, the device shall be returned to the conditioning chamber for not less than 1 hour before shock tests are resumed.

4.4.7 Temperature shock. The device samples shall be subjected to the temperature shock test of MIL-STD-810, Method 503.1, Procedure I, except the high temperature shall be +74°C and the low temperature shall be -48°C.

4.4.8 Functional. The device samples shall be installed in the test fixture, Drawing SA2875117. Each device shall then be subjected to a lanyard pull of 200 ±10 pounds with the safety flag installed. The safety flag shall be removed and the lanyard extracted at a rate of 20 ±2 feet per second with a pull force of 25 ±3 pounds. Failure of one or more of the devices in the sample to meet the requirements of 3.2.1.1 or 3.2.1.2 shall cause rejection of the lot represented.

4.4.9 Lanyard extraction. With the safety pin removed, a static load of 5 pounds shall be applied to the tail fin lanyard. The spool shall freely rotate but not more than 0.5 revolution. Return the spool to its original position indicated in Drawing X4900606 and reinstall the safety pin. Failure of any device to meet the requirements of 3.2.1.3 shall be cause for rejection of the lot represented.

4.5 Packaging, packing, and marking. Prior to shipment, examination shall be made to assure that packaging, packing, and marking conform to Section 5 of this specification.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102. The device shall be packaged in a manner to ensure delivery to the first destination without damage to the device and shall be in accordance with the Code of Federal Regulations, 40 CFR 171-178.

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5.2 Packing. Packing shall be Level C in accordance with FED-STD-102. The package device shall be packed in containers in a manner consistent with safe and good commercial practices complying with the Code of Federal Regulations, 49 CFR 171-178.

5.3 Marking. In addition to any required special markings (see 6.2.1), shipping containers and palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The device is intended to open the tail fins of a binary chemical bomb.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 15 firing devices as specified in 3.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

c. Responsibility for inspection and inspection facilities if different than 4.1.

d. Special markings, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423

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incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited as follows:

Paragraph	Data Requirement	Applicable DID
3.5, 3.3.2.1, 4.1.2.2, 4.2.4	Certificate of compliance	DI-E-2121
3.5, 4.1.2.2, 4.2.4	Inspection report	DI-T-2072

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Safety precautions. The loading, assembly, and handling of the explosives, subassemblies, and the finished items covered by this specification involve hazardous operations and require explosive safety precautions as stated in DoD 4145.26M.

Preparing activity
NAVY - AS

(Project 1377-N700)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-F-85328(AS), FIRING DEVICE, EXPLOSIVE ACTUATOR (CCU-64/B)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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